Refraction

antocciphysics | TEACHER KEY | CPA

Learning target: Connect a change in wave speed to wavelength and direction.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 2; 3 course days later.*
5 cycles; frequency 1/3 Hz ≈ 0.3333 Hz.

02. Explain a worked example (worked)
Refracted angle to normal smaller in the slower medium, for ordinary positive-index media.

03. Look and explain (model)
Toward normal; refracted angle smaller.

04. Build the solution (practice)
4 m and 2 m.

05. Use the idea (reason)
70° from normal.

06. Choose a method (mixed) *Review: Unit 5, Day 5; 12 course days later.*
No. Equilibrium must lie between 14°C and 56°C under the assumptions. Check readings or an unmodeled input.

07. Exit check (exit)
Away from normal, for transmitted ray below critical conditions.

08. STRETCH: Apply it another way (stretch)
Normal incidence remains straight; oblique incidence bends toward normal. Speed changes in both.

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Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 2; 3 course days later.*
$\lambda = 0.3$ m; $f = 1/T \approx 20$ Hz.

02. Explain a worked example (worked)
Refracted angle to normal smaller in the slower medium, for ordinary positive-index media.

03. Represent and interpret (model)
Toward normal; refracted angle smaller.

04. Solve and justify (practice)
$\theta_2 = \sin^{-1}(1/3) = 19.5°$.

05. Analyze the reasoning (reason)
70° from normal.

06. Choose a method (mixed) *Review: Unit 5, Day 5; 12 course days later.*
Water gains 25200 J; metal $c \approx 1050$ J/(kg·°C). Equal energy loss/gain applies under the assumptions.

07. Exit check (exit)
41.8°; total internal reflection, not an ordinary refracted ray.

08. STRETCH: Apply it another way (stretch)
$\theta_2 \approx 19.5°$; reverse critical angle $\sin^{-1}(1/1.5) \approx 41.8°$.